Energy Code Compliance Report

Simulated Building Performance alternative · 2021 IECC Section R405 · Proposed design compared with the Standard Reference Design (Table R405.4.2(1))

SAMPLE DOCUMENT – Fictitious house. Design conditions, construction data and equipment performance are illustrative values, not customer data. Results come from a simplified hourly single-zone model with a synthetic weather year. Code submittals are produced with software approved by the code official and site weather data.

COMPLIES – annual energy cost 57% below the standard reference design

Proposed design $1,160 per year · Standard reference design $2,713 per year · electricity at $0.19/kWh (Illustrative residential electricity price – use approved source per code)

Project	Sample Residence – New Construction
Address	123 Example Lane, Morristown, NJ 07960 (fictitious)
Builder	Example Builders LLC (fictitious)
Building	Two-story single-family home, 4 bedrooms, 2,400 sq ft conditioned floor area, slab-on-grade, vented attic. All-electric: ducted cold-climate heat pump, ERV, heat pump water heater. Air handler and all ducts inside conditioned space (open-web floor trusses between floors).
Climate	IECC climate zone 5A · Morristown, NJ (Morris County) · synthetic reference year: 5,507 HDD65 / 1,016 CDD65, min 4 °F, max 92 °F
Conditioned floor area / bedrooms	2,400 ft² / 4
Code	2021 IECC (as adopted in New Jersey)
Calculation tool	eco-energieberater sample model (hourly, single zone) – for illustration only
Prepared by	[Designer Name], [e.g. ACCA Manual J/D training · RESNET HERS Rater · ASHRAE BEMP]
Date	11/03/2026 · Rev. 0

6,106 kWh
Proposed design, annual

14,277 kWh
Standard reference design, annual

57%
Lower energy cost than reference

2.5 kWh/ft²
Proposed energy intensity (regulated uses)

Compliance criteria

Requirement	Proposed	Limit / reference	Status
Annual energy cost ≤ standard reference design (R405.3)	$1,160	$2,713	**PASS**
Envelope at or better than 2009 IECC prescriptive levels (R405.2)	wall U-0.043 · ceiling U-0.024 · window U-0.27	wall U-0.057 · ceiling U-0.030 · window U-0.35	**PASS**
Mandatory provisions (Table R405.2)	see inspection checklist, section 4	all listed items	to be verified

SAMPLE

1 Proposed design and standard reference design

Building component	Standard reference design	Proposed design
Exterior walls	U-0.045	U-0.043 · 2x6 @ 16" o.c., R-21 cavity + R-5 continuous exterior insulation
Ceiling	U-0.024	U-0.024 · Ceiling below vented attic, R-60 blown cellulose, raised-heel trusses
Window area	360 ft² (15% of CFA, equal on 4 orientations)	367 ft² as designed
Window U / SHGC	U-0.30 / 0.40	U-0.27 / 0.35
Doors	U-0.30	U-0.20
Slab edge	F-0.52	F-0.52 · Slab-on-grade, R-10 vertical edge insulation 4 ft
Air leakage	3.0 ACH50	2.5 ACH50 (blower door test required)
Mechanical ventilation	110 cfm balanced, no heat recovery, 0.83 W/cfm	110 cfm ERV, 70% sensible recovery, 0.60 W/cfm
Heating / cooling	Air-source heat pump at federal minimum efficiency (SEER2 14.3 / HSPF2 7.5)	Variable-capacity cold-climate air-source heat pump, ducted (2-ton class) · SEER2 18.0 / HSPF2 9.5
Distribution system efficiency	0.88	1.00 · ducts and air handler inside conditioned space
Water heating	Electric storage, 50 gal, federal minimum (UEF 0.92)	Heat pump water heater, 50 gal, UEF 3.5
Hot water use	70 gal/day (30 + 10 × N_br)	same
Thermostat	heating 72 °F / cooling 75 °F	same
Internal gains	17,900 + 23.8 × CFA + 4,104 × N_br Btu/day	same
Interior shade fraction	0.92 − 0.21 × SHGC	same

Reference values as applied in this sample per Table R405.4.2(1); confirm against the code edition adopted by the AHJ.

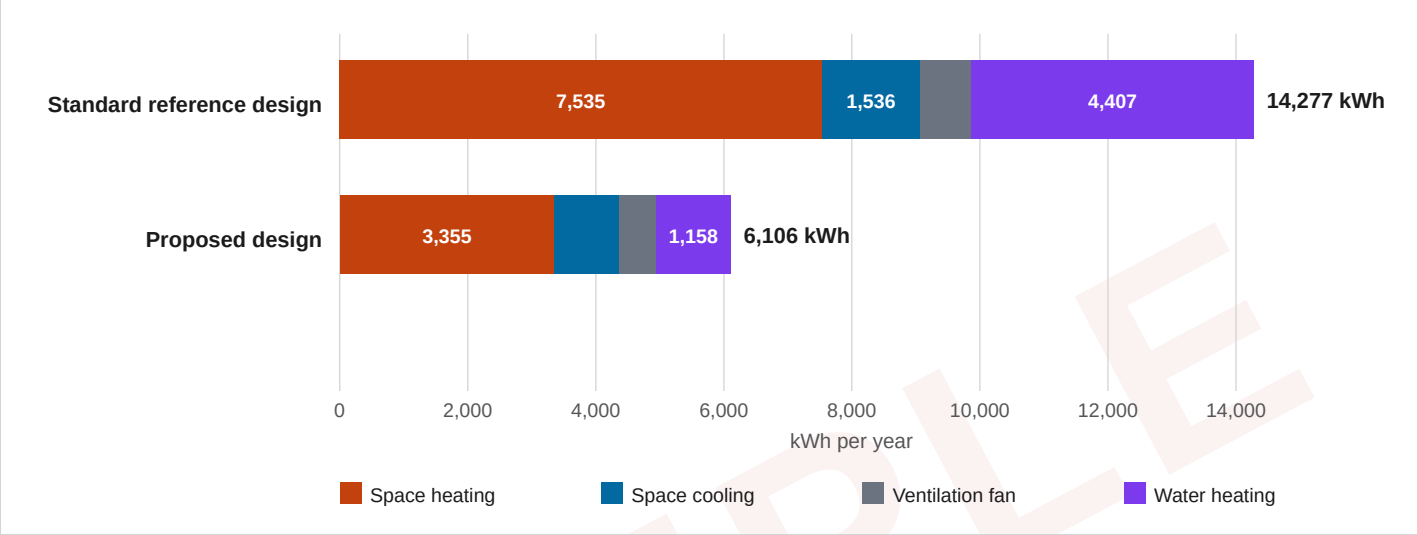
Envelope heat loss coefficient (UA)

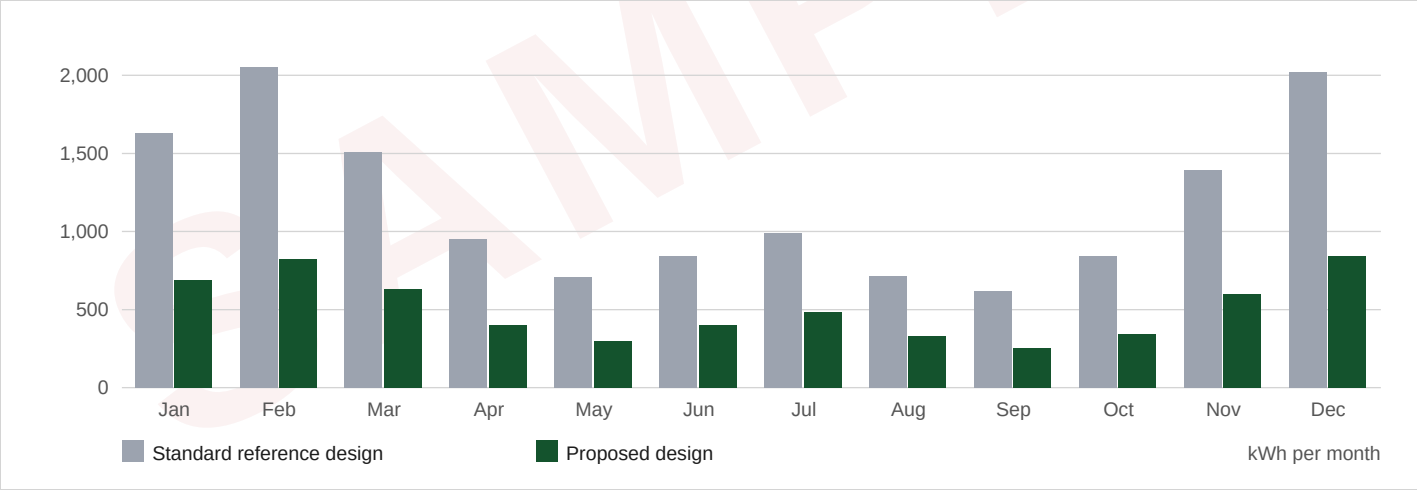
Component	Reference UA Btu/h·°F	Proposed UA Btu/h·°F
Walls	101.7	96.9
Windows	108.0	99.1
Doors	12.0	8.0
Ceiling	28.8	28.8
Slab edge	72.8	72.8
Infiltration	61.2	51.0
Ventilation (net of heat recovery)	118.8	35.6
Total	**503.3**	**392.2**

SAMPLE

2 Annual energy use and cost

End use	Reference kWh	Proposed kWh	Savings kWh	Reference cost	Proposed cost
Space heating	7,535	3,355	4,180	$1,432	$638
Space cooling	1,536	1,014	521	$292	$193
Ventilation fan	800	578	222	$152	$110
Water heating	4,407	1,158	3,249	$837	$220
Total	**14,277**	**6,106**	**8,171**	**$2,713**	**$1,160**





Hours with auxiliary resistance heat: proposed 50 h, reference 1047 h. Lighting and plug loads are identical in both designs and not part of the comparison.

3 Beyond code – upgrade options for the builder

Option (each compared with the proposed design)	Annual kWh	Savings kWh	Savings per year	Cost below reference
Triple-pane windows (U-0.18, SHGC 0.30)	5,653	453	$86	60%
Air sealing to 1.5 ACH50	5,817	290	$55	59%
Walls R-21 + R-10 c.i. (U-0.036)	5,882	225	$43	59%
All three envelope upgrades	5,162	944	$179	64%
All envelope upgrades + rooftop PV 6.0 kW	-2,038 (net)	8,144	$1,547	net producer

PV yield 1,200 kWh per kW (Illustrative yield – verify with NREL PVWatts for the site). Installed costs are not included and are to be priced by the builder. A net-negative result means the home produces more than its regulated uses over a year.

SAMPLE

4 Inspection checklist – component characteristics to verify

Item	Required value	Phase	Verified
Wall insulation	R-21 cavity + R-5 continuous exterior	Insulation inspection	☐
Ceiling insulation	R-60 blown, depth markers, raised-heel trusses	Final inspection	☐
Windows / doors	NFRC labels U ≤ 0.27, SHGC ≤ 0.35; doors U ≤ 0.2	Rough-in	☐
Slab edge insulation	R-10, 4 ft vertical	Foundation	☐
Air leakage (R402.4)	Blower door ≤ 2.5 ACH50 (model input; code maximum 3.0)	Final, test report	☐
Ducts (R403.3)	Air handler and all ducts inside conditioned space, sealed; leakage test as required	Rough-in / final	☐
Equipment sizing (R403.7)	ACCA Manual J / S calculation on file – see HVAC Design Package	Permit	☐
Heat pump	Variable-capacity cold-climate air-source heat pump, ducted (2-ton class), AHRI [AHRI reference no.], SEER2 ≥ 18.0, HSPF2 ≥ 9.5	Final	☐
Controls (R403.1)	Programmable thermostat; aux heat lockout via outdoor sensor	Final	☐
Mechanical ventilation (R403.6)	ERV 110 cfm continuous, balanced, efficacy ≥ 1.2 cfm/W	Final, airflow test	☐
Water heater (R403.5)	Heat pump water heater, 50 gal, UEF 3.5	Final	☐
Hot water pipe insulation (R403.5)	R-3 where required	Rough-in	☐
Lighting (R404)	All permanently installed fixtures high-efficacy	Final	☐

5 Method and limitations

Hourly heat balance of a single thermal zone with an effective building capacitance of 4.0 Btu/°F per ft². Weather: synthetic year with seasonal and daily cycles and random weather variation (fixed seed), solar gains from sun position for each façade. Heat pump capacity and COP from performance tables; load beyond heat pump capacity met by resistance heat. Cooling energy assumes a sensible heat ratio of 0.80. The model shows the structure and content of an R405 report. It is not approved compliance software.

Energy analyst: [Designer Name], [e.g. ACCA Manual J/D training · RESNET HERS Rater · ASHRAE BEMP]
[Company Name] – Remote HVAC & Energy Design · [email]

Signature / date

SAMPLE